

LCG (Linear Congruential Generator) untuk CTF Jeopardy

LCG adalah PRNG paling sederhana dan sering muncul pada challenge crypto pemula-menengah karena mudah dibuat tetapi juga mudah diserang jika parameter atau output bocor.

1. Cara Kerja LCG

Rumus dasarnya:

$$X_{n+1} = (aX_n + c) \bmod m$$

Keterangan:

- X_n = state saat ini
 - a = multiplier
 - c = increment
 - m = modulus
 - X_0 = seed awal
-

Contoh Manual

Misalkan:

$$a = 5$$

$$c = 3$$

$$m = 16$$

$$\text{seed} = 7$$

State pertama:

$$X_1 = (5 \times 7 + 3) \bmod 16$$

$$X_1 = 38 \bmod 16$$

$$X_1 = 6$$

State kedua:

$$X_2 = (5 \times 6 + 3) \bmod 16$$

$$X_2 = 33 \bmod 16$$

$$X_2 = 1$$

State ketiga:

$$X_3 = (5 \times 1 + 3) \bmod 16$$

$$X_3 = 8$$

Output:

6
1
8
11
10
...

2. Challenge CTF Easy

Source Challenge

```
from Crypto.Util.number import *
```

```
flag = b"FLAG{LCG_ATTACK}"
```

```
m_flag = bytes_to_long(flag)
```

```
a = 1103515245
```

```
c = 12345
```

```
mod = 2**31
```

```
seed = 1337
```

```
x1 = (a*seed+c)%mod
```

```
x2 = (a*x1+c)%mod
```

```
x3 = (a*x2+c)%mod
```

```
secret = (a*x3+c)%mod
```

```
enc = m_flag ^ secret
```

```
print("x1 =",x1)
```

```
print("x2 =",x2)
```

```
print("x3 =",x3)
print("enc =",enc)
```

Informasi yang Diberikan ke Peserta

```
x1 = 78628734
x2 = 1460962527
x3 = 934458668
```

```
enc = 23984723984723
```

Analisis

Karena:

```
secret = (a*x3+c)%mod
```

dan semua parameter diketahui.

Peserta tinggal menghitung:

```
secret
```

lalu:

```
flag = enc XOR secret
```

Solver

```
from Crypto.Util.number import *
```

```
a = 1103515245
```

```
c = 12345
```

```
mod = 2**31
```

```
x3 = 934458668
```

```
secret = (a*x3+c)%mod
```

```
enc = 23984723984723
```

```
flag = long_to_bytes(enc ^ secret)
```

```
print(flag)
```

3. Challenge Medium

Sekarang parameter tidak diberikan.

Source Challenge

```
from Crypto.Util.number import *
```

```
a = 1234567
```

```
c = 987654
```

```
m = 2147483647
```

```
seed = 12345
```

```
x1 = (a*seed+c)%m
```

```
x2 = (a*x1+c)%m
```

```
x3 = (a*x2+c)%m
```

```
x4 = (a*x3+c)%m
```

```
print(x1)
```

```
print(x2)
```

```
print(x3)
```

```
print(x4)
```

Peserta hanya mendapat:

x1

x2

x3

x4

Tujuan

Cari:

a

c

m

Ide Matematis

LCG memenuhi:

$$x_2 = a \cdot x_1 + c \bmod m$$

$$x_3 = a \cdot x_2 + c \bmod m$$

Kurangkan:

$$x_3 - x_2 = a(x_2 - x_1) \bmod m$$

dan

$$x_4 - x_3 = a(x_3 - x_2) \bmod m$$

Misalkan:

$$d_1 = x_2 - x_1$$

$$d_2 = x_3 - x_2$$

$$d_3 = x_4 - x_3$$

Maka:

$$d_2 = a \cdot d_1 \bmod m$$

$$d_3 = a \cdot d_2 \bmod m$$

Recover Modulus

Bagian paling terkenal pada attack LCG.

Buat:

$$t_0 = d_2 \cdot d_2 - d_1 \cdot d_3$$

Nilai ini selalu kelipatan modulus.

Jika punya banyak output:

t_0

t1

t2

...

maka:

$$m = \gcd(t_0, t_1, t_2, \dots)$$

Recover a

Setelah modulus diketahui:

$$d_2 = a * d_1 \bmod m$$

maka:

$$a = d_2 * \text{inverse}(d_1, m) \bmod m$$

Recover c

Gunakan:

$$x_2 = a * x_1 + c \bmod m$$

jadi:

$$c = x_2 - a * x_1 \bmod m$$

Recover Semua Output Berikutnya

Setelah mendapat:

a

c

m

semua output masa depan bisa diprediksi.

Challenge Hard

Biasanya flag disembunyikan seperti:

$$\text{keystream} = \text{lcg_output}$$

$$\text{cipher} = \text{flag XOR keystream}$$

Source

```
from Crypto.Util.number import *
```

```
flag = b"FLAG{SECRET}"
```

```
a = 1234567
```

```
c = 987654
```

```
m = 2147483647
```

```
state = 12345
```

```
outputs = []
```

```
for i in range(5):
```

```
    state = (a*state+c)%m
```

```
    outputs.append(state)
```

```
key = state
```

```
enc = bytes_to_long(flag) ^ key
```

```
print(outputs)
```

```
print(enc)
```

Yang Diberikan

```
[1234,5678,....]
```

```
enc=....
```

Langkah Solve

Step 1

Recover:

a

c

m

dari output.

Step 2

Hitung state berikutnya.

$\text{next_state} = (a * \text{last} + c) \% m$

Step 3

Decrypt.

$\text{flag} = \text{enc} \wedge \text{next_state}$

Tools yang Sering Dipakai

Untuk LCG biasanya tidak perlu tool khusus.

Cukup:

```
from math import gcd
```

```
from Crypto.Util.number import inverse
```

Tanda-Tanda Soal LCG

Saat melihat source seperti:

```
state=(a*state+c)%m
```

atau

```
rand=(1103515245*rand+12345)%2**31
```

langsung curiga:

LCG Attack

Karena hampir semua challenge LCG bermuara pada:

1. Recover modulus
2. Recover multiplier
3. Recover increment
4. Prediksi output berikutnya
5. Ambil flag

Itulah pola yang paling sering muncul di CTF Jeopardy kategori Crypto untuk LCG.

